Muhammad Huzaifa

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SUMMARY

Enthusiastic electrical engineering student with four years of hands-on experience in electrical systems and design. Skilled in Kubernetes, machine learning, computer vision, computer and communication networks, and embedded systems. Bilingual in German and English, focused on combining electrical engineering with AI and machine learning to create impactful solutions.

EDUCATION

National University of Sciences and Technology (NUST) — Islamabad, Pakistan
Bachelor of Engineering in Electrical Engineering — November 2021 - June 2025

Army Public School and College (APS&C) — Rawalpindi, Pakistan
Higher Secondary Education, Pre-engineering — April 2019 - July 2021
Percentage: 97.36%

FINAL YEAR RESEARCH PROJECT

Frailty Assessment for the Elderly| IRL, NCAI
Led a fully funded project in collaboration with Juntendo University Graduate School of Medicine, developing a non- contact, computer vision–based frailty assessment system using Kinect V2 and machine learning. Built a user-friendly app for real-time analysis and reporting, achieving up to 100% classification accuracy.

RESEARCH PUBLICATIONS

A Markerless Vision-based Physical Frailty Assessment System for the Elderly

Huzaifa, M*.; Ali, W.; Iqbal, K.F.; Ahmad, I.; Ayaz, Y.; Taimur, H.; Shirayama, Y.; Yuasa, M. *A Markerless Vision-Based Physical Frailty Assessment System for the Older Adults*. AI 2025, 6, 224. **IF: 5.0**

RESEARCH AND HANDS ON EXPERIENCE

Intelligent Field Robotics Lab | Research Intern | NCAI — NUST, Islamabad
June 2024 - June 2025
- Built a non-contact frailty assessment system using Kinect V2 and computer vision.
- Applied ML on joint data for real-time frailty classification.
- Improved model accuracy and deepened expertise in CV-based health analysis.

Adaptive Signal Processing Lab | Research Intern | SINES — NUST, Islamabad
June 2024 - September 2024
- Developed a motor fault detection system using ML, control systems, and cloud integration.
- Used signal processing on sensor data for real-time fault analysis and improved detection accuracy and system reliability

RESEARCH AND INTERNSHIP EXPERIENCE

TUIL LAB| Research Intern | RIMMS — NUST, Islamabad
June 2023 - December 2025
- Study on a 7-element Circularly polarized antenna
- Developing and designing the fabrication of single-element, four
- and 7-element antennae via HFSS Ansys
- Isolation between elements and higher gain for antennas

Optical Networking and Technologies Lab | Research Intern | SINES — NUST, Islamabad
June 2023 - September 2023
- Successfully implementing network communication protocols via Cisco Packet Tracer and open-source network technologies.
- Researched and analysed communication protocols among the network nodes.
- Enhancing skills in network design, protocol analysis

ByteWise Summer Internship | Machine Learning Intern — Islamabad
June 2024 - September 2024
- Worked as a Machine Learning Intern for 100 days, implementing algorithms and models for complex problem solving.

PROJECTS

Face and Eye Detection System using Computer Vision | GitHub
Built a Python-based facial detection engine using pre-trained Haar classifiers with OpenCV, featuring a GUI created in tkinter for real-time face and eye detection.

Plant Disease Identification using Deep Learning | GitHub
A deep learning project using MobileNetV2 and Grad-CAM to classify and visualize plant leaf diseases from the PlantVillage dataset with approximately 90% accuracy.

Fault Detection of Induction Motor using Machine Learning | GitHub
Developed an ML-based system to detect induction motor faults by analyzing sensor data for anomalies like rotor, bearing, and winding faults.

Real-Time SPWM-Based Induction Motor Speed Control via STM32 | GitHub
Designed and implemented a sinusoidal PWM control system in MATLAB/Simulink for precise speed control of a three-phase induction motor integrated with the power grid, ultimately implementing it using STM32 in real-time.

Real-time Image Manipulation in MATLAB | GitHub
Built an image processing system in MATLAB featuring fusion, filtering, edge detection, and enhancement using spatial and frequency domains.

CKD Diagnosis using Machine Learning | GitHub
A machine learning project for diagnosing Chronic Kidney Disease (CKD) by classifying patients using algorithms like Logistic Regression, SVM, and Random Forest.

RTOS-Driven Smart Environment Controller on ESP32 | GitHub
Smart home automation system using the integration of multiple sensors and ESP32 microcontroller

Design of AM Transmitter and Superheterodyne AM Receiver | GitHub
Simulated and validated an AM transmitter and superheterodyne receiver for medium-wave radio using Proteus. Demonstrated modulation and demodulation through signal waveform visualization and real-time system behavior.

Customer Churn Prediction | GitHub
Developed a customer churn prediction model using the Telco Customer Churn dataset, involving data preprocessing, visualization, model training with RandomForestClassifier, and saving the model for future use.

Brain Tumor Detection Using CNNs | GitHub
Developed a deep learning model using convolutional neural networks to classify brain tumors from MRI scans. Achieved high accuracy through image preprocessing, data augmentation, and model optimization.

Real-Time OCR with Deep Learning | GitHub
Built a deep learning-based OCR system using EasyOCR and Python to extract text from live camera feed. Ensured robustness by testing under extreme lighting and poor visual conditions.

CERTIFICATIONS

AI in Healthcare Specialization| Certification
Certified in AI in Healthcare, demonstrating expertise in applying artificial intelligence technologies within the healthcare sector.

Introduction to Machine Learning | Certification
This course offers essential knowledge in machine learning, highlighting important algorithms and techniques. It explores practical applications and provides a certification upon completion to recognize your skills.

Applied AI with DeepLearning | Certification
This course provides hands-on experience in utilizing deep learning techniques to tackle real-world challenges.

Introduction to Tensorflow for AI for Machine Learning and Deep Learning | Certification
This course provides hands-on experience in utilizing deep learning techniques to tackle real-world challenges.

MATLAB OnRamp | Certification
Fundamentals and the basics of MATLAB programming, including data manipulation, visualization, and scripting.

Introduction to Artificial Intelligence (AI) | Certification
This beginner-level course by IBM covers AI basics, applications, and ethics, featuring hands-on labs and a project.

TECHNICAL SKILLS

- **Software Programming Languages:** C/C++, MATLAB, Java, Python
- **Hardware Description Languages:** Verilog and VHDL
- **APIs and Libraries:** Pytorch, Pandas, numpy, Tensorflow, OpenCV, Kubernetes, Docker, Minikube
- **Tools:** MATLAB, Adobe suite, Office suite, Proteus studios, Altium Designer, Visual studios
- **Microcontrollers:** STM32, ESP32, ATMEGA, Arduino, Raspberry pi

LANGUAGES

- English (Fluent) | IELTS 8 Bands
- German (Beginner)
- Urdu (Native/Fluent)

VOLUNTEER EXPERIENCE

- Social Volunteer | Pehli Kiran School, Islamabad
- Social Welfare Volunteer | Red Crescent Society of Pakistan

RELATED COURSEWORK

- Digital Signal Processing
- Signals and Systems
- Computer and Communication Networks
- Engineering Network Analysis
- Machine Learning
- Electrical Machines
- Power Systems Analysis
- Power Systems Protection